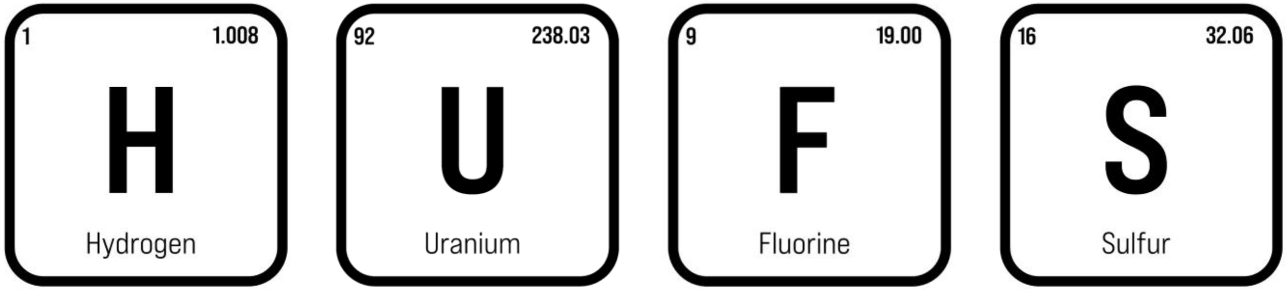
01st Sep 2026



수소경제와 에너지저장 시스템

(Hydrogen Economy and Energy Storage Systems)

Haewon Seo, Ph. D

Hydrogen **E**nerg**Y** Lab **ON** **E**arth

Assistant Professor

Division of Climate Change

Hankuk University of Foreign Studies (HUFS)

Course Introduction

- **Lecture: Hydrogen Economy and Energy Storage Systems (L01212101)**
  - Classroom: 0413
  - Class time: Tuesday, 12:00–14:50

- **Professor**: Haewon Seo
  - Office: Prof Res Bldg , Rm 202
  - E–mail: heyone@hufs.ac.kr
  - Office hour: Monday (till 15:00)

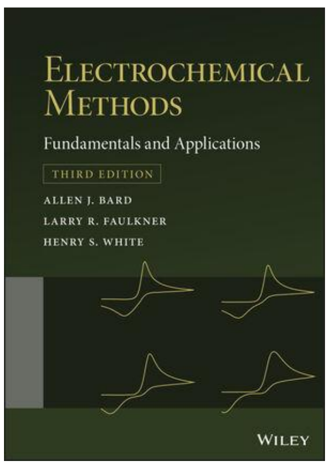
- **Scope**
  - This course introduces hydrogen energy and its associated devices for a clean and sustainable future. Based on fundamental electrochemistry, it examines the operating principles of energy conversion systems for hydrogen production and utilization. Furthermore, it reviews the latest technological trends and explores the development of next-generation energy conversion technologies.

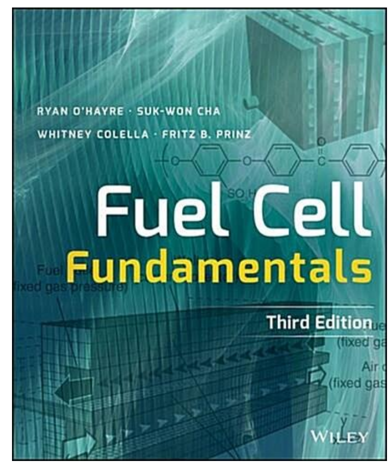
Course Introduction

❑ Textbook

– Main

- Electrochemical Methods (A.J. Bard et al.), 3rd Ed., Wiley (2022)
- Thermodynamics (Y.A. Cengel et al.), 9th Ed., McGraw-Hill (2018)
- Fuel Cell Fundamentals (R. O’Hayer et al.), 3rd Ed., Wiley (2016)





– Reference

- Scientific articles
- IRENA, IEA, World Energy Council, DOE, etc.

Course Introduction

❑ **Schedule**; to be flexibly operated

Week	Date	Contents
01st	**Sep 01st 2026**	**Introduction to Hydrogen (1)**
02nd	Sep 08th 2026	Introduction to Hydrogen (2)
03rd	Sep 15th 2026	Introduction to Hydrogen (3)
04th	Sep 22nd 2026	Fundamentals of Electrochemistry
05th	Sep 29th 2026	Thermodynamics of Reacting Systems
06th	Oct 06th 2026	Electrochemical Cells & Potentials
07th	Oct 13th 2026	Reference Electrodes & Practical Cells
08th	Oct 20th 2026	Midterm exam
09th	Oct 27th 2026	Cell Polarization & Overpotentials
10th	Nov 03rd 2026	Electrode Kinetics & Interface (EDL)
11th	Nov 10th 2026	Mass Transport & Chronoamperometry
12th	Nov 17th 2026	Voltammetry & Reaction Mechanisms
13th	Nov 24th 2026	Electrochemical Impedance
14th	Dec 01st 2026	Applications
15th	Dec 15th 2026	Final exam

Course Introduction

❑ Evaluation

- Attendance (10%)
- Assignment (20%)
- Midterm Exam (30%)
- Final Exam (40%)

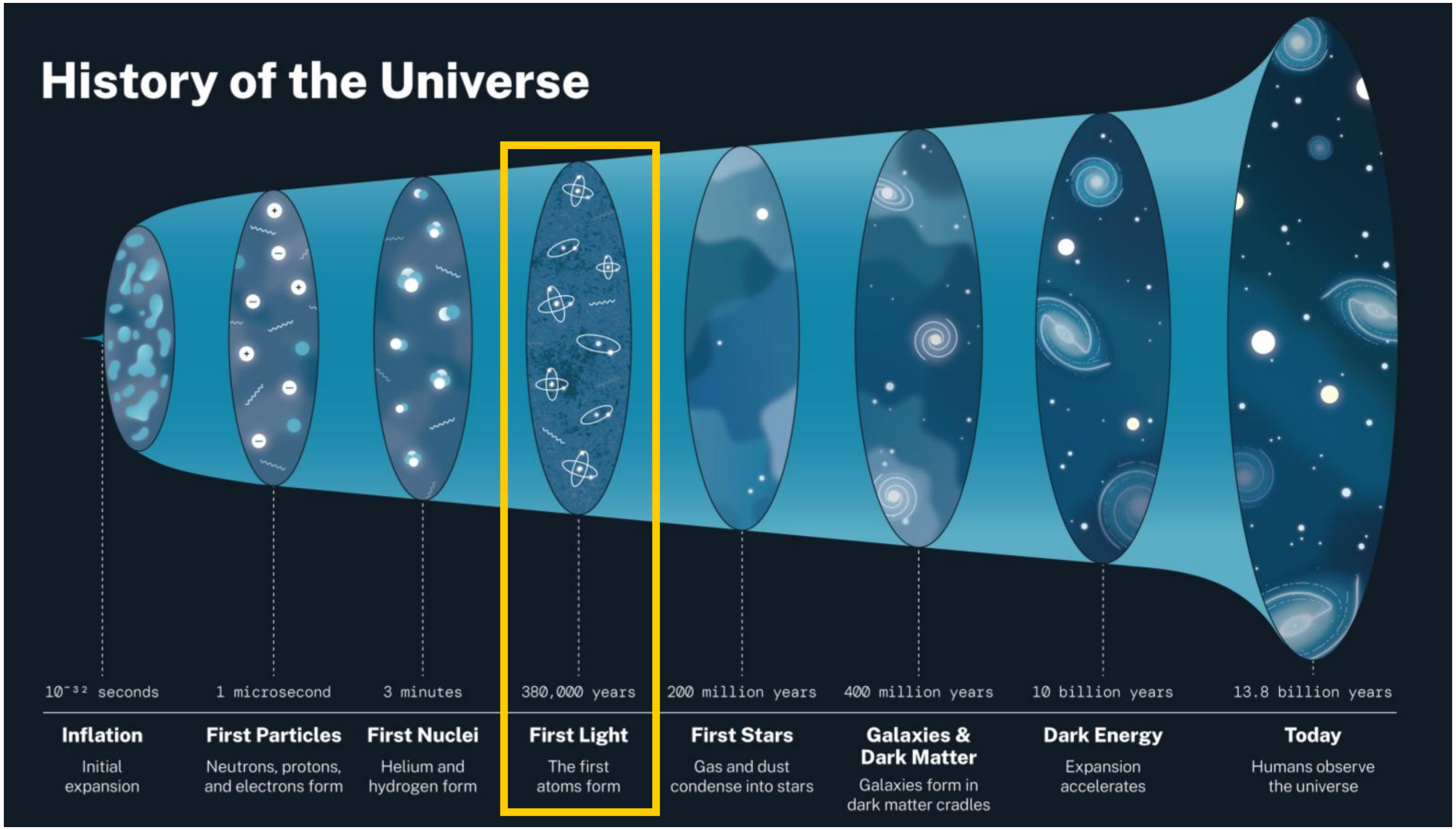
1. Introduction to Hydrogen (1)

Contents

Overview of Hydrogen

https://www.mentimeter.com/app/presentation/alffegrpxtrmxwbokrtq7n7fe4et78v2/present?question=9oz5vx96zvy6

Overview of Hydrogen



– 10^{-35}–10^{-5} s (Hadron epoch): Formation of protons & neutrons (Quark coupling)

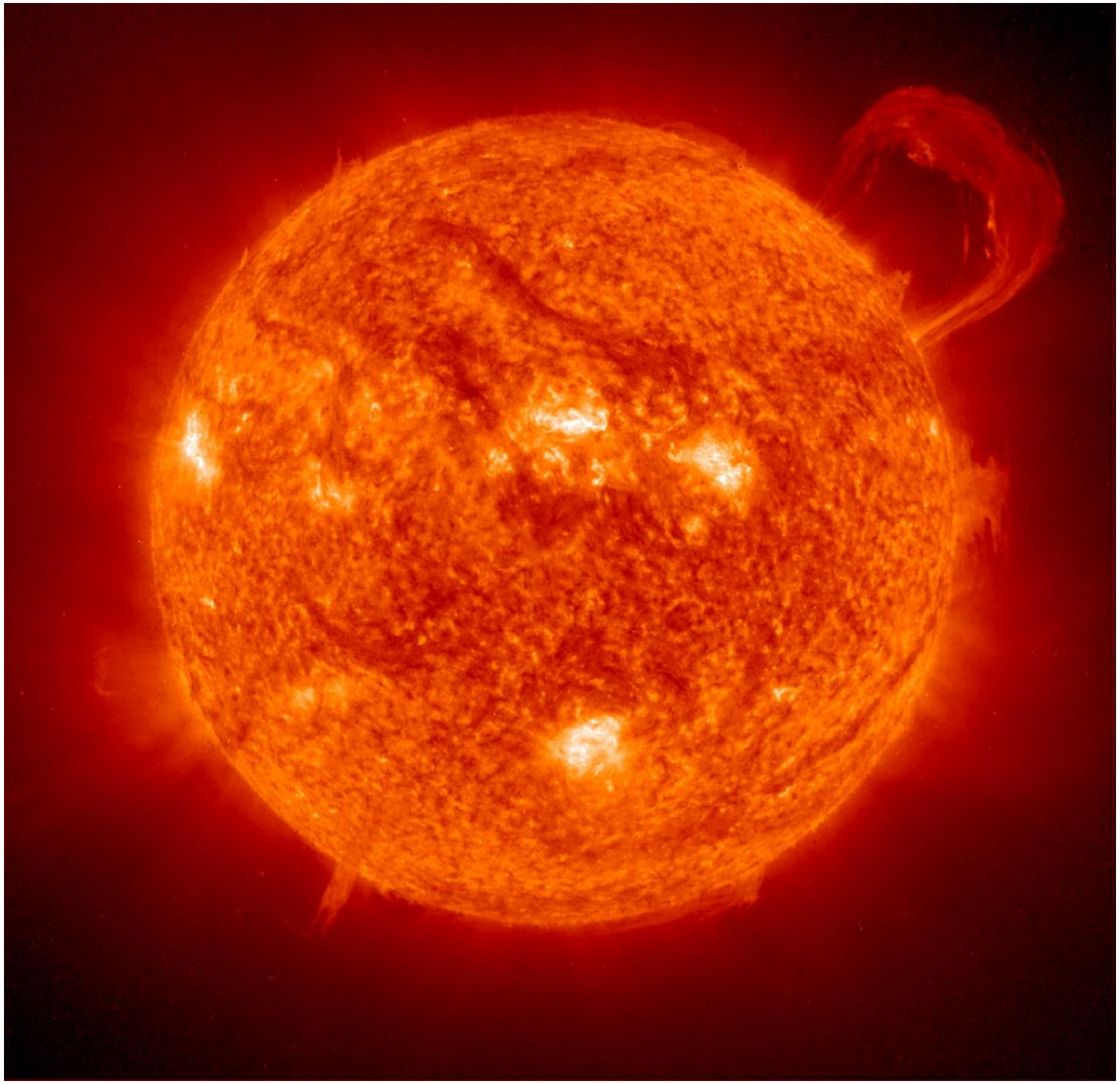
– 10^{-5}–10^{0} s (Lepton epoch): Temperature drop due to further expansion

– 10^{0}–10^{2} s: Formation of atomic nuclei (Nucleosynthesis)

– 10^{2} s–380,000 yr: Mass ratio shifts (Proton/Neutron = 14:2 → H/He = 75:25 wt%)

– After 380,000 yr: Temperature drop and Formation of atom

Overview of Hydrogen

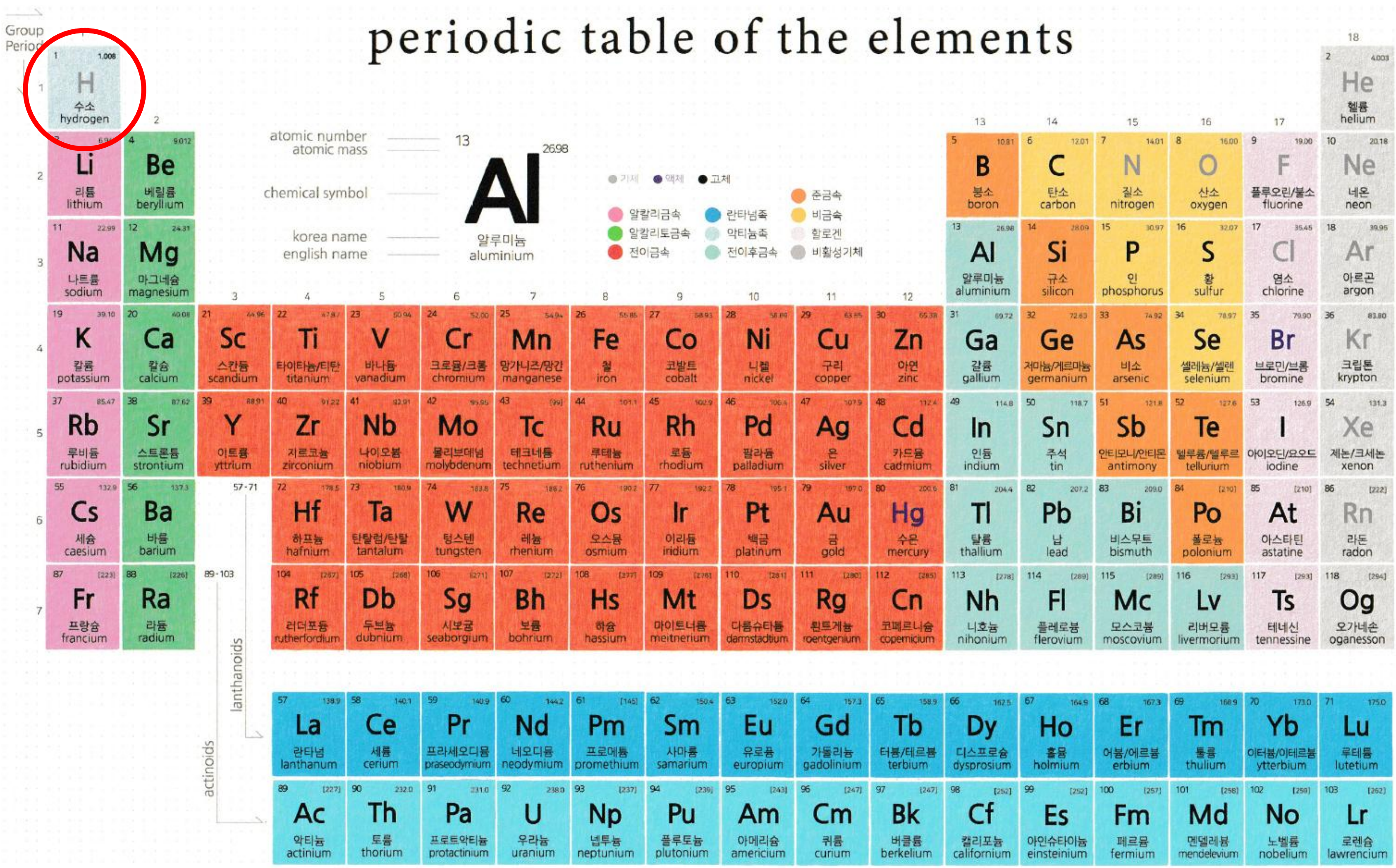


<Source: NASA>

Overview of Hydrogen

❑ Hydrogen

The simplest element in the universe (one proton, one electron, and no neutron)

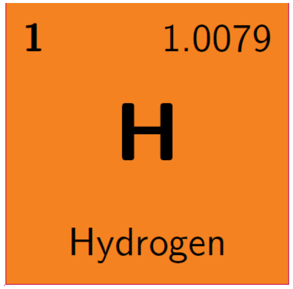


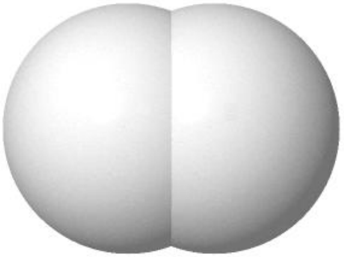
Overview of Hydrogen

❑ Hydrogen

The simplest (one proton & one electron) and abundant (~75%) element in the universe

- Symbol: H
- Atomic Number: 1
- Atomic Weight: 1.0079 u
- Melting Point: –259.2 °C
- Boiling Point: –252.9 °C
- Density: 0.090 g L^{-1} (at STP (0 °C, 1 atm))





–Typically, hydrogen refers to *hydrogen gas* (H_2), which is a colorless, odorless, and nontoxic gas

Table Physical properties of gases at standard temperature and pressure (STP) (0 °C, 1 atm)

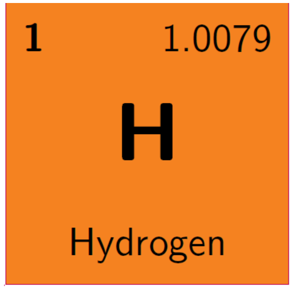
Gas	Density, ρ (g L^{-1})	Specific heat capacity, C_p (J g^{-1} K^{-1})	Thermal cond., k (W m^{-1} K^{-1})	Viscosity, μ (μPa s)
H_2	**0.090**	**14.28**	**0.168**	**8.4**
He	0.179	5.19	0.142	18.6
CH_4	0.717	2.22	0.030	10.3
Air	1.292	1.005	0.024	17.2
O_2	1.429	0.915	0.024	19.2
CO_2	1.977	0.823	0.015	13.7

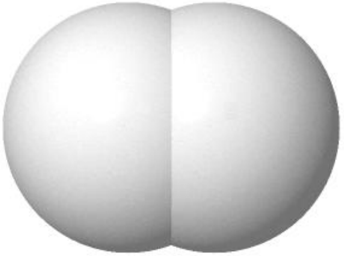
Overview of Hydrogen

❑ Hydrogen

The simplest (one proton & one electron) and abundant (~75%) element in the universe

- Symbol: H
- Atomic Number: 1
- Atomic Weight: 1.0079 u
- Melting Point: –259.2 °C
- Boiling Point: –252.9 °C
- Density: 0.090 g L^{-1} (at STP (0 °C, 1 atm))





–Typically, hydrogen refers to *hydrogen gas* (H_2), which is a colorless, odorless, and nontoxic gas

Table Physical properties of liquids

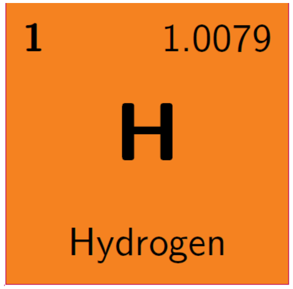
Liquid	Density, ρ (g L^{-1})	Boiling point (°C)	Remarks
H_2	**70.85**	**–252.9**	**~1/6 of CH4's density**
He	125.0	–268.9	
CH_4	422.8	–161.5	
Air	874.0	–194.3	
O_2	1141	–183.0	
CO_2	1178	–78.5	At triple–point

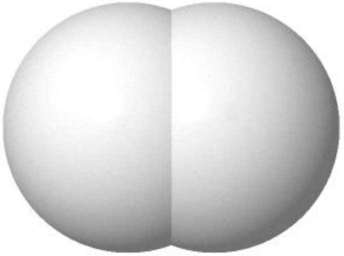
Overview of Hydrogen

❑ Hydrogen

The simplest (one proton & one electron) and abundant (~75%) element in the universe

- Symbol: H
- Atomic Number: 1
- Atomic Weight: 1.0079 u
- Melting Point: –259.2 °C
- Boiling Point: –252.9 °C
- Density: 0.090 g L^{-1} (at STP (0 °C, 1 atm))





–H_2 makes up ~0.5–0.6 ppm of the Earth's atmosphere, whereas H is ubiquitous in water (H_2O), organic compounds (CHONPS), and other matter.

Table Major constituents of the atmosphere of Earth in parts–per million (ppm)

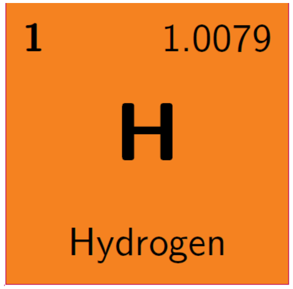
Gas	Volume fraction (ppm)	Gas	Volume fraction (ppm)	Gas	Volume fraction (ppm)
N_2	780,840	CH_4	1.92	O_3	0.07
O_2	209,460	Kr	1.14	NO_2	0.02
Ar	9,340	**H_2**	**0.55**	H_2O	Up to 40,000
CO_2	420	N_2O	0.33		
Ne	18.182	CO	0.10		
He	5.24	Xe	0.09		

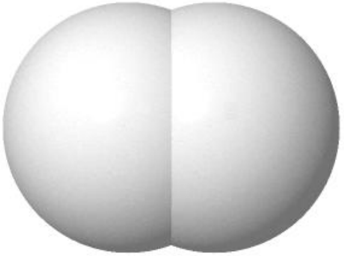
Overview of Hydrogen

❑ Hydrogen

The simplest (one proton & one electron) and abundant (~75%) element in the universe

- Symbol: H
- Atomic Number: 1
- Atomic Weight: 1.0079 u
- Melting Point: –259.2 °C
- Boiling Point: –252.9 °C
- Density: 0.090 g L^{-1} (at STP (0 °C, 1 atm))

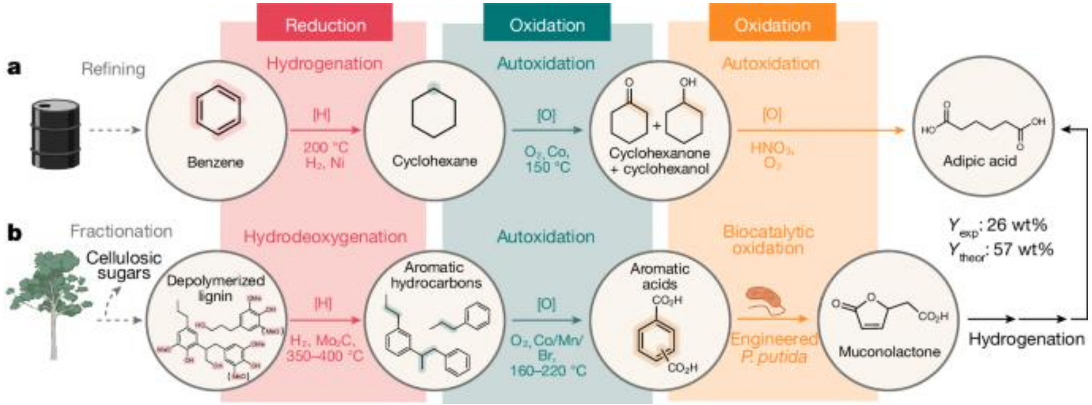




*$H_2 \rightarrow 2H$ ($\Delta H°$ = 435.14 kJ mol^{-1})

–Low reactivity at room temperature, but reacts with most elements and compounds at high temperatures or in the presence of catalysts

–Involved in biological redox (oxidation–reduction) reactions



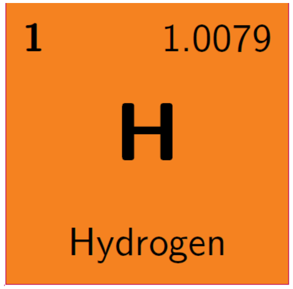
Nature, **654** (2026) 668–675

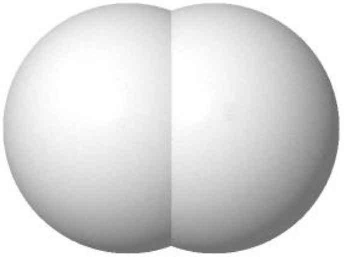
Overview of Hydrogen

❑ Hydrogen

The simplest (one proton & one electron) and abundant (~75%) element in the universe

- Symbol: H
- Atomic Number: 1
- Atomic Weight: 1.0079 u
- Melting Point: –259.2 °C
- Boiling Point: –252.9 °C
- Density: 0.090 g L^{-1} (at STP (0 °C, 1 atm))





–Reacts with halogens to form hydrogen halides, and with CO to produce hydrocarbons and alcohols

*$H_2 + X_2 \rightarrow 2HX$ (X = F, Cl, Br, I)

**$nCO + (2n+1)H_2 \rightarrow C_nH_{2n+2} + nH_2O$

***$nCO + 2nH_2 \rightarrow C_nH_{2n+1}OH + (n-1)H_2O$

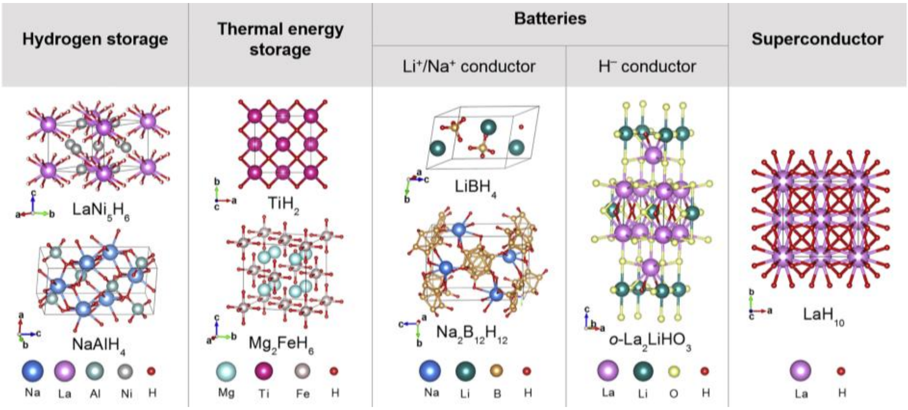
–Combines with numerous elements to form various hydrides

*Ionic hydride (groups 1, 2 except for Be, Mg) (e.g., LiH, NaH, CaH_2, etc.)

**Covalent hydride (groups 13–17) (e.g., B_2H_6, CH_4, NH_3, H_2O, HF, etc.)

***Metallic hydride (typically transition metals) (e.g., TiH_2, PdH_x, etc.)

***Complex hydride (e.g., $NaBH_4$, $LiAlH_4$, $NaAlH_4$, etc.)



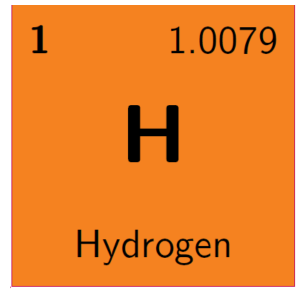
Joule, **4**(4) (2020) 705–709

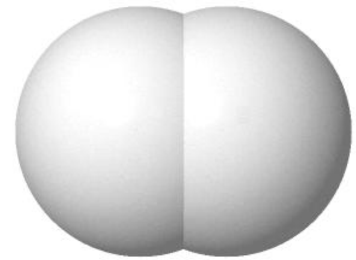
Overview of Hydrogen

❑ Hydrogen

The simplest (one proton & one electron) and abundant (~75%) element in the universe

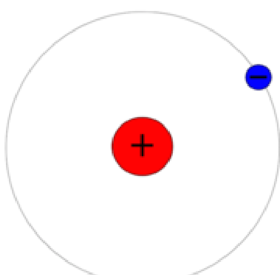
- Symbol: H
- Atomic Number: 1
- Atomic Weight: 1.0079 u
- Melting Point: –259.2 °C
- Boiling Point: –252.9 °C
- Density: 0.090 g L^{-1} (at STP (0 °C, 1 atm))



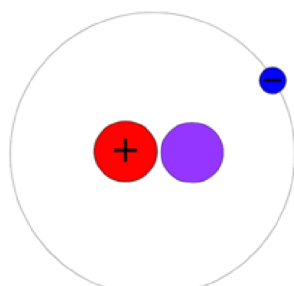


–Hydrogen has three main naturally occurring isotopes: ^{1}H (protium), ^{2}H (D) (deuterium), ^{3}H (T) (tritium)

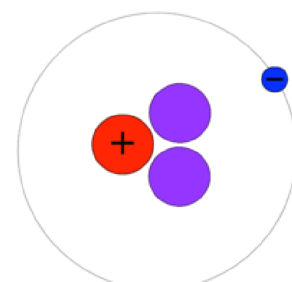
Protium



Deuterium



Tritium



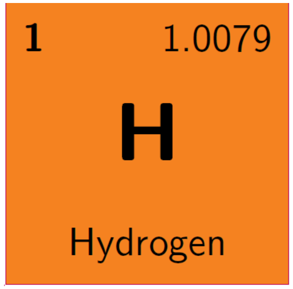
	Protium	Deuterium	Tritium
Atomic Weight	1.0079 u	2.0141 u	3.0160 u
Half-Year	Stable	Stable	12.32 yr
Abundance	99.9855%	0.0115%	trace

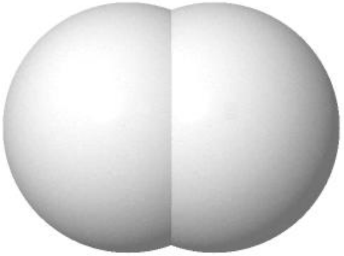
Overview of Hydrogen

❑ Hydrogen

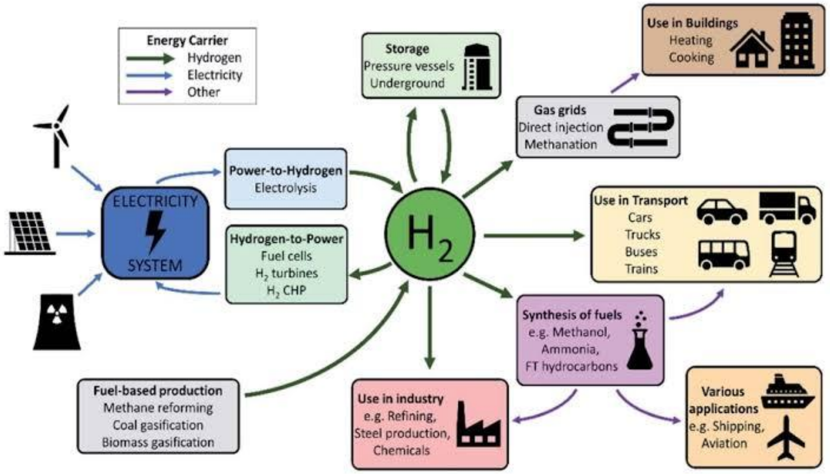
The simplest (one proton & one electron) and abundant (~75%) element in the universe

- Symbol: H
- Atomic Number: 1
- Atomic Weight: 1.0079 u
- Melting Point: –259.2 °C
- Boiling Point: –252.9 °C
- Density: 0.090 g L^{-1} (at STP (0 °C, 1 atm))





–H_2 is *an energy carrier* that is converted from primary energy sources into a secondary form, making it **storable** and **transportable**.

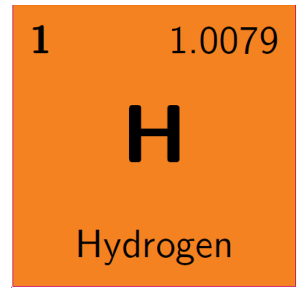


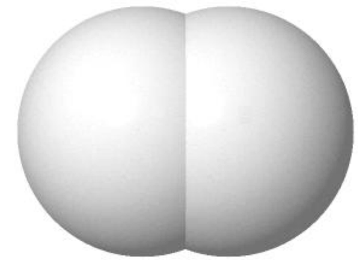
Overview of Hydrogen

❑ Hydrogen

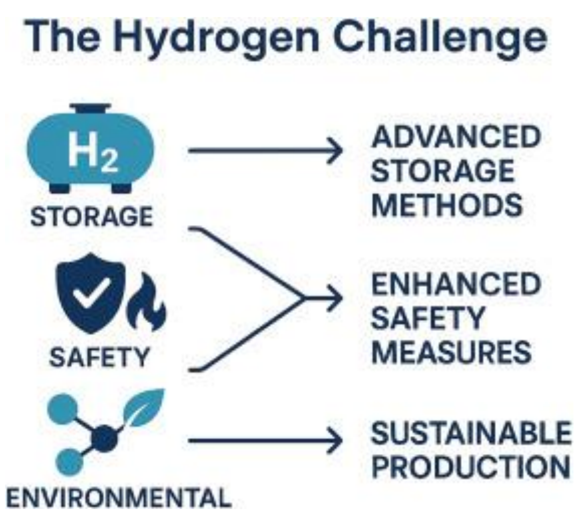
The simplest (one proton & one electron) and abundant (~75%) element in the universe

- Symbol: H
- Atomic Number: 1
- Atomic Weight: 1.0079 u
- Melting Point: –259.2 °C
- Boiling Point: –252.9 °C
- Density: 0.090 g L^{-1} (at STP (0 °C, 1 atm))

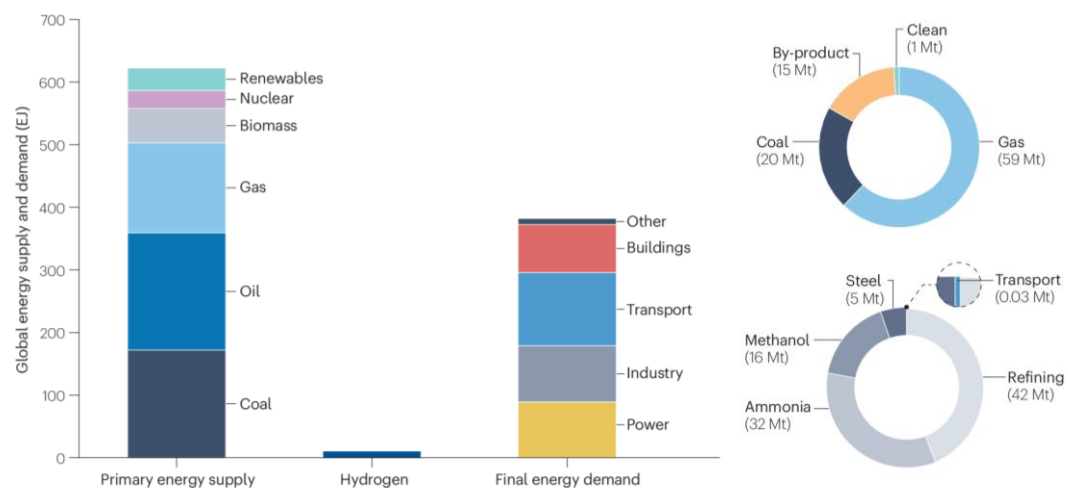




–Major challenges include production costs, storage limitations, the development of highly efficient combustion engines, and safety; **the economic viability needs be improved**.



Int. J. Hydrogen Energy, **167** (2025) 150952



Nat. Rev. Clean Technol., **1** (2025) 351–371

Quiz

❑ Why Is Hydrogen So Scarce in Earth's Atmosphere?

H_2 makes up only about 0.55 ppm of Earth's atmosphere (See Slide #11). Even though Earth is known as the "water planet," atmospheric H_2 is extremely scarce. A common explanation often states:
“Because H_2 is the lightest molecule, it continually rises through the atmosphere due to buoyancy and eventually escapes into outer space.”
Is this explanation physically accurate? If not, what is the correct scientific mechanism behind the low concentration and escape of H_2 from Earth's atmosphere?

(Hint 1)
A helium balloon floats because surrounding air pushes it up. But what happens to a balloon when there is no air left to push it? Can "buoyancy" kick something entirely out into space?

(Hint 2)
Below 100 km, the Earth's atmosphere is constantly stirred by strong winds and turbulence. Does H_2 really rise to the top by itself?

(Hint 3)
Consider the Maxwell–Boltzmann distribution for root-mean-square speed ($v_{rms} = 3k_bT/m)^{0.5}$ vs. Earth’s escape velocity v_{esc} = 11.2 km s^{-1} (k_b: Boltzmann constant, T: Temperature, m: mass)

End of Document